

Supplementary Mathematical Appendix

A theory of templating substrates

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This appendix provides formal definitions and proofs for the mathematical claims in the main paper. The material is organized into five sections corresponding to the apparatus, mode-specific information bounds, descriptor-relativity, the formal sense of bulk-matched controls, and the capacity and generation theorems with their bounded-heredity corollary and finite-population proposition.

The appendix is written for a reader with graduate-level information theory ([Cover and Thomas, 2006](#)) and basic familiarity with discrete memoryless channels. Notation follows the main paper; $|\mathcal{A}|$ denotes alphabet cardinality, $h_b(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ is binary entropy, and $D_{\text{KL}}(P \parallel Q)$ is Kullback-Leibler divergence.

Contents

1	Formal apparatus definitions	2
1.1	The templating event	2
1.2	Population mutual information	2
1.3	Substrate capacity	3
1.4	Channel mutual information	3
1.5	Per-base mechanism fidelity	4
1.6	The apparatus signature	4
2	Mode-specific information bounds	4
2.1	Mode 1 — linear scaling	4
2.2	Mode 3 — logarithmic saturation	5
2.3	Mode 5 — module-bounded	5
2.4	Mode 2 — code-mediated capacity ceiling	6
2.5	Mode 4 — length independence	6
2.6	Mode 6 — bounded by surface positions, parasitic on Mode 1	7
3	Descriptor-relativity	7
4	Bulk-matched controls as null hypotheses	8
5	Capacity and generation theorems	8
5.1	Notation and structural conditions	9
5.2	Heredity capacity	9
5.3	Deterministic closure and cumulative generation	10
5.4	Capacity theorem	11
5.5	Generation theorem	11
5.6	Bounded-heredity corollary	12
5.7	Finite-population proposition	13

1 Formal apparatus definitions

1.1 The templating event

A templating event is a tuple $\mathcal{E} = (X, O, S, \Delta G; \varphi_X, \varphi_Y, Y)$ where:

- $X \in \mathcal{X}$ is the template, drawn from a template population with distribution Π_X (which may be a point mass δ_x for a single fixed template).
- O is the operator (separable molecular machinery; possibly empty if templating is autocatalytic).
- S is the substrate pool from which Y is constructed.
- ΔG is the free-energy budget driving the templating reaction.
- $\varphi_X : \mathcal{X} \rightarrow \mathcal{X}'$ and $\varphi_Y : \mathcal{Y} \rightarrow \mathcal{Y}'$ are descriptor maps on template and product spaces.
- Y is the structured product, distributed as $Y \mid (X, O, S, \Delta G) \sim P_{\text{chan}}(\cdot \mid X)$, where P_{chan} is the templating channel induced by O acting on $(X, S, \Delta G)$.

The templating channel $P_{\text{chan}}(\cdot \mid X)$ is a stochastic kernel from $\mathcal{X}$ to $\mathcal{Y}$. When $\varphi_X = \varphi_Y$ the templating is *same-descriptor*; otherwise it is *cross-descriptor* and O implements the descriptor coupling.

1.2 Population mutual information

Definition 1 (Population MI). For a templating event $\mathcal{E}$ with template population $\mathcal{X}$ and distribution Π_X ,

$$I_{\text{struct}}^{\text{pop}}(\mathcal{E}) := H(\varphi_Y(Y)) - \mathbb{E}_{X \sim \Pi_X} [H(\varphi_Y(Y) \mid X)]$$

where the marginal $H(\varphi_Y(Y))$ is computed under the law $Y \sim \int_{\mathcal{X}} P_{\text{chan}}(\cdot \mid x) d\Pi_X(x)$ and the conditional entropies are under $P_{\text{chan}}(\cdot \mid X)$.

Proposition 2 (Population MI bound). $I_{\text{struct}}^{\text{pop}}(\mathcal{E}) \leq H(\Pi_X)$.

Proof. $I_{\text{struct}}^{\text{pop}}$ is the mutual information $I(X; Y)$ computed under Π_X and the channel P_{chan} . By the data-processing inequality applied to φ_Y as deterministic post-processing of Y , $I(\varphi_X(X); \varphi_Y(Y)) \leq I(X; Y) \leq H(X) = H(\Pi_X)$. $\square$

Corollary 3 (Fixed-template degeneracy). When $\mathcal{X} = \{x_0\}$ is a single fixed template, $H(\Pi_X) = 0$, so $I_{\text{struct}}^{\text{pop}}(\mathcal{E}) = 0$ by construction.

This corollary is the formal content of the framework’s claim that biological systems with single fixed templates (Drt3a’s single ncRNA, a single chromosome being replicated within a cell) have zero population MI. Population MI measures *cross-realization information transfer*; for a fixed template, there are no realizations to transfer between.

1.3 Substrate capacity

Definition 4 (Substrate capacity). For a templating substrate class $\mathcal{S}$ and template length L , the capacity is

$$C_{\mathcal{S}}(L) := \sup_{\Pi_X \text{ on } \mathcal{X}_L} I_{\text{struct}}^{\text{pop}}(\mathcal{E}).$$

The supremum is over template-distribution choices on the substrate's L -length template space $\mathcal{X}_L$. $C_{\mathcal{S}}$ is a property of the substrate class, not of any particular biological realization. In contrast, $I_{\text{struct}}^{\text{pop}}$ is the realized population MI in a specified system. Both are needed; conflating them is the most common failure mode of substrate-level reasoning.

Remark 5. For Mode 1 (Watson-Crick) at alphabet size $|\mathcal{A}|$ and per-position misincorporation rate ε , $C_{\mathcal{S}}(L) = L \cdot [\log_2 |\mathcal{A}| - h_b(\varepsilon) - \varepsilon \log_2 (|\mathcal{A}| - 1)]$, achieved by the uniform distribution Π_X^{unif} (Theorem 12). For Mode 3 with N -state cyclic dynamics, $C_{\mathcal{S}}(L) \leq \log_2 N$ for all L , regardless of template-distribution choice (Theorem 13).

1.4 Channel mutual information

Definition 6 (Channel MI against an ensemble). Let $\mathcal{C} = \{c_1, \dots, c_K\}$ be a finite ensemble of templating channels with prior $P(C)$. For a focal channel $c \in \mathcal{C}$,

$$I_{\text{struct}}^{\text{chan}}(c; \mathcal{C}) := D_{\text{KL}}(P(\varphi_Y(Y) \mid C = c) \parallel \bar{P}(\varphi_Y(Y)))$$

where the mixture distribution is

$$\bar{P}(\varphi_Y(Y)) := \sum_{c' \in \mathcal{C}} P(C = c') P(\varphi_Y(Y) \mid C = c').$$

The full ensemble channel MI is

$$\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) := \sum_{c \in \mathcal{C}} P(C = c) D_{\text{KL}}(P(\varphi_Y(Y) \mid C = c) \parallel \bar{P}).$$

Proposition 7 (Channel MI bound). $\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) \leq \log_2 |\mathcal{C}|$, with equality iff the channels in $\mathcal{C}$ produce mutually disjoint output distributions.

Proof. $\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) = I(C; Y)$ under the joint law $P(C, Y) = P(C)P(Y \mid C)$. $H(C) \leq \log_2 |\mathcal{C}|$. The mutual information is bounded by either marginal entropy. Equality of $I(C; Y)$ and $H(C)$ occurs iff Y determines C , i.e., iff the conditional distributions $P(Y \mid C = c)$ have disjoint supports. $\square$

Remark 8 (Channel MI for fixed templates). Unlike $I_{\text{struct}}^{\text{pop}}$, $I_{\text{struct}}^{\text{chan}}$ does not vanish for fixed-template channels. A channel with a single fixed template still produces a distinguishable output distribution against alternative channels, and the divergence $D_{\text{KL}}(P(Y \mid C = c) \parallel \bar{P})$ measures that distinguishability. This is the formal content of the framework's claim that $I_{\text{struct}}^{\text{chan}}$ distinguishes Drt3a (Mode 1 with fixed ncRNA template) from Drt3b (Mode 3 with cyclic active site) even when both are individual cellular realizations with fixed mechanisms.

1.5 Per-base mechanism fidelity

Definition 9 (Phase fidelity and monomer fidelity). For a templating channel with cyclic dynamics having N states, define a phase-trajectory function $\sigma : \mathbb{N} \rightarrow \{0, 1, \dots, N-1\}$ that gives the canonical phase for each position. The *phase fidelity* is

$$f_{\text{phase}}(c) := \mathbb{E} \left[\frac{1}{L} \sum_{i=1}^L \mathbf{1}\{\hat{\sigma}_i(Y) = \sigma(i)\} \right]$$

where $\hat{\sigma}_i$ is the phase recovered from position i of Y . The *monomer fidelity* at state s is

$$f_{\text{monomer}}(c; s) := \mathbb{E} \left[\frac{1}{|\{i : \sigma(i) = s\}|} \sum_{i: \sigma(i)=s} \mathbf{1}\{Y_i = \pi(s)\} \right]$$

where $\pi(s)$ is the canonical monomer for state s . For acyclic channels (Mode 1, Mode 5), $N = 1$, $f_{\text{phase}} \equiv 1$, and only f_{monomer} is meaningful, simplifying to the per-position correctness rate.

Remark 10. f_{phase} and f_{monomer} are not mutual informations. They are moments of the joint distribution capturing different mechanism-level observables. The split is necessary because cyclic channels can fail in two structurally distinct ways: an architectural disruption that breaks f_{phase} (the cycle no longer alternates) versus a selectivity disruption that degrades f_{monomer} within an intact cycle (the alternation continues but with degraded monomer choice). These correspond to the architectural-versus-selectivity gate distinction reported in main paper Results.

1.6 The apparatus signature

Definition 11 (Joint signature). The apparatus signature for a templating channel c against an ensemble $\mathcal{C}$ is the tuple

$$\sigma(c; \mathcal{C}) := \left(I_{\text{struct}}^{\text{pop}}(c), I_{\text{struct}}^{\text{chan}}(c; \mathcal{C}), f_{\text{phase}}(c), f_{\text{monomer}}(c; s) \right)_{s=0}^{N-1}.$$

Mode classification uses the joint signature; signature collapse to any single coordinate is information-discarding.

2 Mode-specific information bounds

This section proves the scaling-law claims for each of the five primary modes. Mode 6 is treated separately in §2.6.

2.1 Mode 1 — linear scaling

Theorem 12 (Mode 1 capacity). For a Watson-Crick channel with alphabet $|\mathcal{A}| \geq 2$, per-position misincorporation rate ε , template population $\mathcal{X} = \mathcal{A}^L$ with uniform distribution Π_X^{unif} , and product Y of length L ,

$$I_{\text{struct}}^{\text{pop}}(X; Y) = L \cdot [\log_2 |\mathcal{A}| - h_b(\varepsilon) - \varepsilon \log_2 (|\mathcal{A}| - 1)].$$

Proof. The Watson-Crick channel is discrete memoryless: $P(Y_i = \text{wc}(x_i) \mid X_i = x_i) = 1 - \varepsilon$, and $P(Y_i = a' \mid X_i = x_i) = \varepsilon / (|\mathcal{A}| - 1)$ for each $a' \neq \text{wc}(x_i)$. By memorylessness, the channel decomposes per position. For any position i , $I(X_i; Y_i) = H(X_i) - H(X_i \mid Y_i)$. Under uniform Π_X , $H(X_i) = \log_2 |\mathcal{A}|$. By Bayes' rule and uniformity, $H(X_i \mid Y_i) = H(Y_i \mid X_i) = h_b(\varepsilon) + \varepsilon \log_2 (|\mathcal{A}| - 1)$. Summing over L positions gives the stated formula. $\square$

Empirical match. Across 30 sweep cells with $\varepsilon \in \{0.001, 0.01, 0.05, 0.10, 0.25\}$ and $L \in \{10, 25, 50, 100, 200, 500\}$, $n_{\text{samples}} = 5000$ per cell, the empirical $I_{\text{struct}}^{\text{pop}}$ matches Theorem 12 within 0.6% across all cells (Test A.1, main paper Fig. 1A).

2.2 Mode 3 — logarithmic saturation

Theorem 13 (Mode 3 capacity bound). *For an N -state cyclic templating channel with phase $X \in \{0, 1, \dots, N-1\}$ drawn uniformly, and product Y of length L generated by deterministic cycle progression with per-step substrate-selection error ε ,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq \log_2 N$$

for all L . The bound is approached in the low-noise long-block limit ($L \rightarrow \infty$ with ε fixed at small positive value); the bound is saturated exactly at $\varepsilon = 0$.

Proof. By the data-processing inequality applied to the cycle progression (which is a deterministic function of phase and time step), $I(X; Y) \leq I(X; \text{phase trajectory}) \leq H(X) = \log_2 N$. The bound is achieved iff phase is recoverable from Y . At $\varepsilon = 0$ the cycle is noise-free and phase is recoverable from Y_1 alone (the first position determines the phase that produced it). At $\varepsilon > 0$ the recoverability is imperfect on a single position but improves with L because the periodic structure of Y provides redundant phase information. Standard channel-coding asymptotic analysis (Cover and Thomas, 2006, Theorem 8.7.1) gives $\lim_{L \rightarrow \infty} I(X; Y) = \log_2 N$ for fixed $\varepsilon > 0$ in the regime where the cycle's per-step Shannon capacity exceeds zero. $\square$

Empirical match. For $N \in \{2, 3, 4, 5, 6, 8, 10\}$, $\varepsilon = 10^{-3}$, L from $L = N$ to $L = 50N$: empirical $I_{\text{struct}}^{\text{pop}}$ approaches $\log_2 N$ to within 0.2% across all tested cells (worst case at $N = 2$, $L = 2$; bound is $< 0.01\%$ for $N \geq 3$; Test B, main paper Fig. 2A,B).

2.3 Mode 5 — module-bounded

Theorem 14 (Mode 5 capacity). *For a modular conveyor with N modules, each independently selecting one monomer from substrate alphabet of size k with fidelity $1 - \varepsilon$, and product Y of length L :*

$$\begin{aligned} I_{\text{struct}}^{\text{pop}}(X; Y) &= N \cdot [\log_2 k - h_b(\varepsilon) - \varepsilon \log_2(k-1)] & \text{for } L \geq N, \\ I_{\text{struct}}^{\text{pop}}(X; Y) &= L \cdot [\log_2 k - h_b(\varepsilon) - \varepsilon \log_2(k-1)] & \text{for } L < N, \end{aligned}$$

with positions $L > N$ (when $L > N$) carrying zero per-position information.

Proof. Each module independently determines one product position; modules 1 through $\min(L, N)$ are templated, positions $N + 1, \dots, L$ (when $L > N$) have no module association and are drawn from a fixed distribution independent of the modular template. Per-position MI follows the same memoryless argument as Theorem 12 for templated positions and equals 0 for non-templated positions. $\square$

Empirical match. Test C, main paper Fig. 2D.

2.4 Mode 2 — code-mediated capacity ceiling

Theorem 15 (Mode 2 capacity bound). *For code-mediated translation with DNA template of L codons drawn uniformly random from the 64 codons, standard genetic code (NCBI table 1), translation error rate $\varepsilon \ll 1$, and peptide output Y of length L over 20 amino acids plus stop,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq L \cdot H(\nu),$$

where ν is the codon-count distribution: $\nu(a) = n(a)/64$, with $n(a)$ the number of codons mapping to amino acid a in the standard genetic code (e.g., $n(\text{Leu}) = 6$, $n(\text{Met}) = 1$).

Proof. $I(X; Y) \leq H(Y)$. For uniform random X , the marginal distribution of Y_i (per codon) is exactly ν , so $H(Y_i) = H(\nu)$. By memorylessness of per-codon translation, $H(Y) = L \cdot H(\nu)$. $\square$

Empirical match. $H(\nu) = 4.218$ bits/codon for the standard genetic code (computed from the codon table, including stop as a 21st symbol); empirical $I_{\text{struct}}^{\text{pop}}$ at $L = 64$, $n_{\text{samples}} = 5000$, gives 4.214 bits/codon (Test F4). Match to four significant figures.

Capacity comparison with Mode 1. The same DNA at length $3L$ nucleotides has Mode 1 capacity $6L$ bits (Theorem 12 at $|\mathcal{A}| = 4$). Mode 2 capacity is $L \cdot H(\nu) = 4.218L$ bits. Capacity ratio $= 6/4.218 \approx 1.42$. Code degeneracy reduces information transfer by approximately 30% from the substrate ceiling.

Note on alternative bounds. The frequently-cited $L \cdot \log_2(20) = 4.32L$ bits assumes uniform marginal over amino acids, which the standard genetic code does not produce; the correct bound uses $H(\nu)$. Including stop as a 21st symbol gives $H(\nu) = 4.218$ bits; restricting to the 61 sense codons only gives $H(\nu_{\text{sense}}) \approx 4.139$ bits.

2.5 Mode 4 — length independence

Theorem 16 (Mode 4 capacity bound). *For a conformer-state templating channel with K distinguishable conformers and templated polymers of length L ,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq \log_2 K$$

independently of L .

Proof. X (the templating conformer) takes K values; $H(X) \leq \log_2 K$. The product Y inherits the conformer assignment as a single discrete attribute (its conformational state), regardless of polymer length. The polymer’s primary structure is fixed by the genome, not by the conformer. $\square$

Remark 17. Mode 4 is “state-templating” rather than “position-templating”; the length-independence is a property of the substrate (conformational identity), not a degenerate special case of position templating. The framework treats Mode 4 as $O(1)$ information content as a structural feature, not as an asymptotic limit.

2.6 Mode 6 — bounded by surface positions, parasitic on Mode 1

For a 2D surface-position templating system with N distinguishable surface positions and per-position selection from k options, the combinatorial capacity bound is

$$C_S^{\text{combinatorial}}(N) \leq N \cdot \log_2 k,$$

which can in principle scale unboundedly as the surface area grows. Mode 6 capacity scaling is therefore not bounded by dimensionality alone.

The framework’s distinguishing claim about Mode 6 is that biological instances fail two operational requirements distinct from raw combinatorial capacity:

1. **Parasitic copyability.** The operator that copies the surface pattern, and the units that the surface assembles, are both produced by Mode 1 acting on the genome. Failure of Mode 1 inheritance terminates Mode 6 inheritance. Mode 6’s apparent capacity is gated by Mode 1’s substrate.
2. **Bounded heritable variation.** The variation that biological Mode 6 transmits is geometric perturbation of an existing unit repertoire, not compositional novelty. The number of distinguishable cortical states a lineage can transmit is bounded by combinatorial arrangements of genome-encoded parts, not by the surface’s combinatorial position count. Open-ended scaling requires unboundedly many heritable distinguishable states; biological Mode 6 lacks this because new compositional novelty would require new genome-encoded parts.

We omit a formal bound because Mode 6’s biological instances (ciliate cortex) are not fully characterized at the apparatus level in the published literature. The operational implementation of bounded heredity in the H3 mechanism characterization is the empirically relevant claim; see §5 below.

3 Descriptor-relativity

Definition 18. Two descriptor maps φ_Y, φ'_Y on $\mathcal{Y}$ are *equivalent* if there exists a bijection π such that $\varphi'_Y = \pi \circ \varphi_Y$. A templating event $\mathcal{E}$ analyzed under non-equivalent descriptors φ_Y and φ'_Y may give different $I_{\text{struct}}^{\text{pop}}$ values.

Theorem 19 (Descriptor non-uniqueness for amyloid templating). *Let an amyloid β -sheet polymer A propagate its conformational state to a peptide P . Under the descriptor $\varphi_Y^{\text{seq}} = \text{“amino-acid sequence of } P\text{,”}$ the templating event classifies as Mode-1-like with information bound determined by the β -sheet residue-pairing constraints. Under the descriptor $\varphi_Y^{\text{conf}} = \text{“}\beta\text{-sheet conformational state of } P\text{,”}$ the same templating event classifies as Mode 4 with information bound $O(1)$.*

Sketch. Under φ_Y^{seq} , the joint distribution of P ’s amino-acid sequence given A ’s amino-acid sequence has the structure of a position-by-position channel constrained by β -sheet pairing rules; $I_{\text{struct}}^{\text{pop}}$ scales linearly with L subject to those constraints. Under φ_Y^{conf} , only the conformational state is observed; P ’s conformation inherits A ’s regardless of length, so $I_{\text{struct}}^{\text{pop}}$ is bounded by $\log_2(\text{number of distinguishable conformations}) = O(1)$. $\square$

Corollary 20. *Mode classification applies to (templating event, descriptor) pairs, not to physical objects. The same template can mediate templating events that classify into different modes under different descriptors.*

This is what allows the framework to avoid an unresolvable taxonomic question of the form “is amyloid templating Mode 1 or Mode 4?” The question is ill-posed; the answer depends on what is being measured.

4 Bulk-matched controls as null hypotheses

Definition 21 (k -th order matched null). Given a product Y from a templating event, the k -th order matched null distribution Q_k is the maximum-entropy distribution on $\mathcal{Y}$ that preserves all k -th order marginal statistics of the empirical distribution of Y . Specifically: Q_1 preserves single-position marginals (composition) and is iid from those marginals; Q_2 preserves nearest-neighbor frequencies and is the maximum-entropy first-order Markov chain consistent with those frequencies; Q_k preserves k -position joint marginals and is the maximum-entropy $(k - 1)$ -th order Markov chain.

Remark 22. For $k = 2$, Q_2 is *not* iid; samples from Q_2 are first-order Markov sequences whose nearest-neighbor frequencies match the empirical Y .

Definition 23 (Bulk-matched control). The bulk-matched control for a templating event $\mathcal{E}$ is a templating-free null in which:

- Y_{bulk} is drawn from Q_k for some specified k (typically $k = 1$ or $k = 2$).
- X_{bulk} is drawn independently from any distribution; the templating channel is replaced by independence of X_{bulk} and Y_{bulk} .

$I_{\text{struct}}^{\text{pop}}(X_{\text{bulk}}; Y_{\text{bulk}})$ measures the apparent information that arises purely from k -th order compositional structure shared between template and product, with no position-specific templating.

Proposition 24 (Bulk control as null). *A claim that $\mathcal{E}$ is a templating event with structural information transfer requires*

$$I_{\text{struct}}^{\text{pop}}(X; Y) > \tau \cdot I_{\text{struct}}^{\text{pop}}(X_{\text{bulk}}; Y_{\text{bulk}})$$

where τ is a stated significance threshold (the main paper uses $\tau \geq 20$ for a “pass” verdict).

The threshold τ encodes the strength of the claim. Compositional bias alone can produce non-zero $I_{\text{struct}}^{\text{pop}}$ through k -th order shared statistics; the bulk-matched control isolates position-specific information transfer from the compositional contribution. Whether a k -th order match is a sufficient null depends on what k captures the confounders: for Mode 1 versus random output the bulk-match at $k = 1$ is sufficient; for Mode 3 versus Mode 1 with periodic-like template, $k \geq 2$ is needed because nearest-neighbor structure differs. The framework reports controls with $k = 2$ by default.

5 Capacity and generation theorems

This section formalizes the two necessity results referenced in the main paper: the *capacity theorem* (R1, R2, R3 are necessary for unlimited heredity capacity) and the *generation theorem* (R4 is additionally necessary for cumulative generative recursion beyond deterministic closure). The structure is: notation and structural conditions (§5.1); formal definition of heredity capacity with its upper bound (§5.2); formal definition of deterministic closure and cumulative generative recursion (§5.3); the capacity theorem with proof (§5.4); the generation theorem with proof (§5.5); the bounded-heredity corollary recovering each mode’s capacity ceiling (§5.6); the finite-population, finite-horizon proposition (§5.7); and the recovery of prior formulations within this structure (§5.8).

5.1 Notation and structural conditions

Let $\mathcal{X}_L$ denote the space of admissible configurations at size parameter L ; this is countable in all biological examples we treat. Let Σ be a finite alphabet, called the *recognition alphabet* of the catalytic operation. A descriptor is a map $\varphi_X : \mathcal{X}_L \rightarrow \Sigma^{N(L)}$ for some position count $N(L) \geq 1$. The catalytic operation $\mathcal{O}$ has kernel $Q_L : \mathcal{X}_L \times \mathcal{X}_L \rightarrow [0, 1]$ giving the conditional probability of producing $X' = \mathcal{O}(X)$ from template X :

$$Q_L(X' | X) \geq 0, \quad \sum_{X' \in \mathcal{X}_L} Q_L(X' | X) = 1 \text{ for all } X. \quad (1)$$

We allow $\mathcal{O}$ to be composed with a reversible involution $\iota : \mathcal{X}_L \rightarrow \mathcal{X}_L$ (e.g., Watson–Crick complementation), with the recursion taking the form $X \rightarrow \iota(X') \rightarrow X'' \rightarrow \dots$; for notational simplicity we suppress ι where it does not affect the argument.

The five conditions of the substrate recursion framework are:

- (R1) Recognition-alphabet entropy.** $|\Sigma| \geq 2$, so $\log_2 |\Sigma| > 0$.
- (R2) Unbounded extensible recognition.** $\liminf_{L \rightarrow \infty} N(L) = \infty$, where $N(L)$ is the count of independently variable recognized positions of the descriptor φ_X under Q_L . “Independently variable” is meant in the first-order, descriptor-relative sense: the substrate is not constrained to a finite recurrent phase repertoire, and increasing L increases the count of heritable recognized degrees of freedom. The condition does not require strong joint-statistical independence across positions; context-dependent error rates (as in real DNA polymerases) are permitted.
- (R3) Catalytic closure under iteration.** For every $X \in \mathcal{X}_L$ in the support of Q_L , $\mathcal{O}(X) \in \mathcal{X}_L$ (possibly via the reversible involution ι), so that $\mathcal{O}^2(X)$ is well-defined and the iteration $X \rightarrow \mathcal{O}(X) \rightarrow \mathcal{O}^2(X) \rightarrow \dots$ remains within $\mathcal{X}_L$.
- (R4) Stochastic drift in Σ .** The kernel $Q_L(\cdot | X)$ has $|\text{supp } Q_L(\cdot | X)| \geq 2$ for at least some $X \in \mathcal{X}_L$, with the dispersion of $\varphi_X(X')$ around $\varphi_X(X)$ lying in the recognition alphabet Σ of (R1). Drift in a separate alphabet $\Sigma' \neq \Sigma$ (e.g., chemical modifications outside the polymerase’s recognition palette) fails R4.
- (R5) Scope: intrinsic catalysis.** The kernel Q_L is determined by intrinsic physicochemistry — the active-site free-energy landscape, the available reactants, the chemical driving — not by an externally specified loss function or target product. R5 is scope-defining rather than structural: it demarcates the domain in which R1–R4 apply rather than introducing a fifth necessity condition.

5.2 Heredity capacity

Let Π_X denote any probability distribution on $\mathcal{X}_L$ (an input ensemble of templates), and let $X' \sim Q_L(\cdot | X)$ for $X \sim \Pi_X$.

Definition 25 (Heredity capacity). The *heredity capacity* of $(\mathcal{X}, \mathcal{O})$ at size L is

$$C_S(L) = \sup_{\Pi_X} I(\varphi_X(X); \varphi_X(X')), \quad (2)$$

the supremum of mutual information between input and output descriptors over all input ensembles.

Proposition 26 (Capacity upper bound). *For any kernel Q_L ,*

$$C_S(L) \leq N(L) \cdot \log_2 |\Sigma|. \quad (3)$$

Proof. By the standard identity $I(A; B) = H(B) - H(B | A) \leq H(B)$, for any input ensemble Π_X ,

$$I(\varphi_X(X); \varphi_X(X')) \leq H(\varphi_X(X')).$$

The descriptor $\varphi_X(X')$ takes values in $\Sigma^{N(L)}$, a set of cardinality $|\Sigma|^{N(L)}$, so

$$H(\varphi_X(X')) \leq \log_2 |\Sigma|^{N(L)} = N(L) \log_2 |\Sigma|.$$

The supremum over Π_X inherits the bound. $\square$

The bound is achievable iff some Π_X realizes a uniform $\varphi_X(X')$ on $\Sigma^{N(L)}$. Realizability requires that the recognition step actually resolve all $|\Sigma|$ states at each of $N(L)$ positions and that the iteration close within $\mathcal{X}_L$ — i.e., R1, R2, R3.

Definition 27 (Unlimited heredity capacity). $(\mathcal{X}, \mathcal{O})$ has *unlimited heredity capacity* if $C_S(L) = \Omega(L)$ as $L \rightarrow \infty$.

Remark 28 (Information-theoretic and combinatorial formulations). Definition 27 is the information-theoretic statement. Its combinatorial equivalent: in the discrete noiseless limit where the upper bound of Proposition 26 is saturated, the recursively readable configuration set $\mathcal{R}_L = \{X \in \mathcal{X}_L : \mathcal{O}^t(X) \in \mathcal{X}_L \text{ for all } t \geq 0\}$ satisfies $|\mathcal{R}_L| \geq 2^{cL}$ for some $c > 0$. The two formulations are coordinated through entropy: linear scaling of $C_S(L)$ in bits is equivalent to exponential scaling of the recursively readable state count. They describe the same fact at two levels.

5.3 Deterministic closure and cumulative generation

Definition 29 (Deterministic restriction). A *deterministic restriction* of Q_L is a function $f : \mathcal{X}_L \rightarrow \mathcal{X}_L$ such that $Q_L(f(X) | X) > 0$ for every $X \in \mathcal{X}_L$. Let $\mathcal{D}(Q_L)$ denote the set of all deterministic restrictions of Q_L .

If Q_L is itself deterministic ($|\text{supp } Q_L(\cdot | X)| = 1$ for all X), then $\mathcal{D}(Q_L)$ is a singleton. If Q_L is genuinely stochastic at some state, $|\mathcal{D}(Q_L)| > 1$: each $f \in \mathcal{D}(Q_L)$ selects one positive-probability successor at each branching state.

Definition 30 (Deterministic closure). The *deterministic closure* of a set $S_0 \subset \mathcal{X}_L$ under Q_L is

$$\text{cl}_{\text{det}}(S_0) = \bigcup_{f \in \mathcal{D}(Q_L)} \bigcup_{t \geq 0} f^t(S_0). \quad (4)$$

Equivalently, $\text{cl}_{\text{det}}(S_0)$ is the set of configurations reachable from S_0 by non-branching positive-probability paths.

Definition 31 (Cumulative generative recursion). $(\mathcal{X}, \mathcal{O})$ supports *cumulative generative recursion* if, for every L sufficiently large and for every finite proper subset $S_0 \subsetneq \mathcal{X}_L$ with $|S_0| \ll |\mathcal{R}_L|$, the recursively readable descendant set

$$\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \{X \in \mathcal{X}_L : \exists X_0 \in S_0, t \geq 0 \text{ with } Q_L^t(X | X_0) > 0 \text{ and } \mathcal{O}^s(X) \in \mathcal{X}_L \text{ for all } s \geq 0\} \quad (5)$$

strictly contains $\text{cl}_{\text{det}}(S_0)$, with the additional configurations differing from elements of $\text{cl}_{\text{det}}(S_0)$ in local positions of the descriptor φ_X (i.e., in the alphabet Σ).

The qualifier “strictly contains” is essential and replaces an earlier formulation that required only $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) \not\subset S_0$. That earlier formulation was false as a characterization of R4: a deterministic counter chain $000 \rightarrow 001 \rightarrow 010 \rightarrow \dots$ leaves any singleton S_0 without being stochastic, but the descendant set is then a single predetermined orbit contained in $\text{cl}_{\text{det}}(S_0)$. Definition 31 distinguishes deterministic-orbit traversal (which can leave S_0 but not exceed $\text{cl}_{\text{det}}(S_0)$) from genuine stochastic branching in Σ (which exceeds $\text{cl}_{\text{det}}(S_0)$).

5.4 Capacity theorem

Theorem 32 (Capacity necessity). *Within the (R5) scope, if $(\mathcal{X}, \mathcal{O})$ has unlimited heredity capacity in the sense of Definition 27, then R1, R2, and R3 hold.*

Proof. We prove the contrapositive: failure of any of R1, R2, R3 forces $C_S(L)$ to scale strictly slower than $\Omega(L)$.

Case R1 fails. If $|\Sigma| = 1$, then $\log_2 |\Sigma| = 0$. By Proposition 26, $C_S(L) \leq N(L) \cdot 0 = 0$, so $C_S(L) \equiv 0$, not $\Omega(L)$.

Case R2 fails. If $\liminf_{L \rightarrow \infty} N(L) = N_{\max} < \infty$, then $N(L) \leq N_{\max}$ for all L sufficiently large. By Proposition 26, $C_S(L) \leq N_{\max} \log_2 |\Sigma|$, a constant in L . So $C_S(L) = O(1)$, not $\Omega(L)$.

Case R3 fails. If $\mathcal{O}(X) \notin \mathcal{X}_L$ for some X in the support of Q_L (and not recoverable via reversible involution ι), then $\mathcal{O}^2(X)$ is undefined within $(\mathcal{X}, \mathcal{O})$. The recursion does not iterate within the substrate–operator pair. Any further heredity is transferred to a different substrate class $(\mathcal{X}', \mathcal{O}')$ that itself satisfies R3 (parasitic recursion). The heredity capacity of $(\mathcal{X}, \mathcal{O})$ as a self-contained recursive substrate is bounded by the information transmitted in a single application of $\mathcal{O}$, which is at most $N(L) \log_2 |\Sigma|$, and subsequent rounds are not properly attributable to $(\mathcal{X}, \mathcal{O})$. The $\Omega(L)$ scaling of recursive heredity capacity cannot be attributed to $(\mathcal{X}, \mathcal{O})$ alone.

By contrapositive, $C_S(L) = \Omega(L)$ requires R1, R2, and R3. $\square$

Remark 33 (R4 is not required for capacity). Theorem 32 does not invoke R4. A perfect copier — $Q_L(X' | X) = \delta_{X', X}$, no drift — trivially satisfies $C_S(L) = N(L) \log_2 |\Sigma|$: the channel is the identity on $\mathcal{X}_L$, so $I(\varphi_X(X); \varphi_X(X')) = H(\varphi_X(X))$, and choosing Π_X uniform on $\mathcal{X}_L$ achieves $H(\varphi_X(X)) = N(L) \log_2 |\Sigma|$. This is the structural reason a hypothetical drift-free DNA polymerase would still be a high-capacity inheritance carrier — it would not innovate from a finite seed, but it would store and transmit the full 4^L -fold configurational repertoire faithfully.

5.5 Generation theorem

Theorem 34 (Generation necessity). *Within the (R5) scope, if $(\mathcal{X}, \mathcal{O})$ supports cumulative generative recursion in the sense of Definition 31, then R4 holds, in addition to the R1, R2, R3 inherited from coherence of the catalytic process.*

Proof. R1, R2, R3 are preconditions for cumulative generation as defined: cumulative generation requires a recognition alphabet (R1), at least one extensible position (R2 in the recognition-relevant sense), and a recursion that closes within $\mathcal{X}_L$ (R3). We focus on R4.

Suppose R4 fails. There are two cases.

Case (a): the kernel is deterministic. If $|\text{supp } Q_L(\cdot | X)| = 1$ for all $X \in \mathcal{X}_L$, then $\mathcal{D}(Q_L) = \{f\}$ for the unique function $f(X) =$ the (unique) positive-probability successor of X . By Definition 30, $\text{cl}_{\text{det}}(S_0) = \bigcup_{t \geq 0} f^t(S_0)$. Every positive-probability trajectory of Q_L from S_0 is contained in this orbit closure: at each X , the only positive-probability successor is $f(X)$, so the recursively readable

descendant set is

$$\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \bigcup_{t \geq 0} f^t(S_0) = \text{cl}_{\text{det}}(S_0).$$

This is not strict containment. Cumulative generation in the sense of Definition 31 fails.

Case (b): the kernel drifts in an alphabet $\Sigma' \neq \Sigma$. Suppose $Q_L(X' | X) > 0$ for some $X' \neq X_{\text{ref}}(X)$, but the difference $\varphi_X(X') \neq \varphi_X(X)$ at position i involves an alphabet character $\sigma' \in \Sigma' \setminus \Sigma$. Then $\varphi_X(X')$ is not in $\Sigma^{N(L)}$ under the recognition map φ_X : the drift is invisible to the next iteration’s recognition step. The next $\mathcal{O}$ -application reads $\varphi_X(X')$ as if it lay in $\Sigma^{N(L)}$, treating σ' as the closest Σ -character or as undefined; either way, the drift in Σ' does not propagate as recursively readable variation. Hence X' with Σ' -drift is excluded from $\mathcal{R}_{Q_L}^{\text{rec}}(S_0)$ (the recursive-readability requirement). The descendant set therefore does not strictly contain $\text{cl}_{\text{det}}(S_0)$ *through branching local alternatives in Σ* (Definition 31’s qualifier).

In both cases, the cumulative-generation requirement fails. By contradiction, R4 holds whenever $(\mathcal{X}, \mathcal{O})$ supports cumulative generation. $\square$

Remark 35 (Why deterministic closure is essential). A deterministic counter chain illustrates why the “leaves S_0 ” formulation was insufficient. Take $\mathcal{X}_L = \{0, 1\}^L$, $S_0 = \{0^L\}$, and $\mathcal{O}(X) = X + 1 \bmod 2^L$ (increment in binary). Then for any $T \geq 1$, $\mathcal{O}^T(S_0) = \{T \bmod 2^L\}$, which is outside S_0 . The earlier formulation of cumulative generation (“descendant set leaves S_0 ”) would be satisfied — yet the kernel is deterministic, so R4 fails. The deterministic-closure formulation correctly excludes this case: $\text{cl}_{\text{det}}(S_0) = \{0, 1, \dots, 2^L - 1\}$ already, so $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \text{cl}_{\text{det}}(S_0)$ is not strict containment.

5.6 Bounded-heredity corollary

The capacity and generation theorems give the bounded-heredity classification of substrates failing a strict subset of R1–R4.

Corollary 36 (Bounded heredity, mode-keyed). *Within the (R5) scope, substrates failing a strict subset of R1–R4 exhibit heredity bounded according to which condition fails:*

1. *Failure of R1 at the per-site level ($|\Sigma| = 1$): $C_S(L) = 0$. No heritable distinguishability per site.*
2. *Failure of R1 at the macro level via a finite attractor repertoire of size K , with no extensible recognition substructure: $C_S(L) \leq \log_2 K$, length-independent. Mode 4: prion strains, amyloid attractors, DRT1 filament conformer.*
3. *Failure of R2 via a finite recurrent phase repertoire of N states: $C_S(L) \leq \log_2 N$, regardless of product length. Mode 3: Drt3b at $N = 2$ gives $C_S(L) \leq 1$ bit.*
4. *Failure of R2 via finite module count N with per-module alphabet of size k : $C_S(L) \leq N \log_2 k$. Mode 5: NRPS, polyketide synthases.*
5. *Failure of R3: substrate-level heredity is bounded by the information transmitted in a single application of $\mathcal{O}$; recursion is parasitic on a separate substrate–operator pair that itself satisfies R3. Mode 2 (translation: protein not ribosome-readable; recursion parasitic on Mode 1 acting on the gene encoding the ribosome). Mode 6 (cortex: surface does not autonomously self-copy; recursion parasitic on Mode 1 acting on the genome).*
6. *Failure of R4: substrate has heredity capacity but cannot generate novelty from a finite initial repertoire. Heredity is bounded by the founding-population repertoire.*

The corollary is a direct consequence of the upper bound in Proposition 26 together with the contrapositive structure of Theorems 32–34: each condition failure picks off a specific multiplicative or additive term in the bound. The biological assignment to Modes 1–6 follows the main paper’s taxonomy table.

5.7 Finite-population proposition

The bounded-heredity corollary is a substrate-level statement (the heredity capacity ceiling at the kernel level). The corresponding population-level statement — what happens when a substrate–operator pair with a given failure pattern hosts a finite population under selection — requires careful scoping. The universal-cap formulation (“mechanisms failing R3 or R4 are structurally capped at $F_k^* < F_{\max}$ under any population dynamics”) is too strong, because a wholesale-redraw mechanism with full support over $\mathcal{X}_L$ can sample any configuration by chance with infinite time. The proper finite-horizon, finite-population form follows.

Proposition 37 (Finite-horizon plateau under nonlocal or absent drift). *Let a population of size K evolve under selection sharpness β toward a target phenotype on a tested-class landscape (single-target Hamming distance, L -dimensional alphabet, static), over a horizon $T < \infty$. Then:*

1. *Mechanisms whose substrate–operator pair fails R4 at the individual level (exact-copy systems: M1 lineage-level fixation, M3 individual-level exact copy) cannot exceed the maximum fitness F_0^* present in the founding lineage repertoire. Mean fitness at horizon T is bounded by F_0^* regardless of K , β , or T .*
2. *Mechanisms whose substrate–operator pair has nonlocal R4 (wholesale redraw: M2 at rate $r > 0$) sample novelty with full support over $\mathcal{X}_L$ but do not accumulate local improvements. Under the tested update rules, the probability of reaching the target X^* within horizon T is dominated by random hitting time $\tau \sim |\mathcal{X}_L|/K$, which scales as $\exp(cL)$ for $L \gg 1$ and is negligible at biological scales.*
3. *Mechanisms whose substrate–operator pair satisfies R3 and R4 with individual-level local drift (M4) are not structurally barred from cumulative climbing toward $F_{\max}$. Whether climbing actually occurs depends on landscape topology, finite-population drift, mutation load, neutral-network connectivity, and other conditions outside the framework’s scope.*

The proposition is empirically tested in the main paper’s Test H3–H5 sweep. The tested class ($L_{\text{target}} \in \{16, 32, 64, 128\}$, $K = 400$, $\beta \in \{2, 5, 10, 20\}$, 30 replicates per cell) confirms each item: M0 (no recursion) plateaus at chance (0.250); M1 and M3 (R4 fails) plateau near $F_0^* \approx 0.47$ –0.50; M2 at sweet-spot $r^* = 0.10$ plateaus at ~ 0.53 ; M4 reaches ~ 0.97 . The proposition’s positive direction (M4 not structurally barred) is bounded: the simulation shows climbing on the tested landscape; it does not establish climbing on arbitrary landscapes.

5.8 Synthesis and recovery of prior formulations

The capacity and generation theorems recover and clarify prior formulations of the inheritance question within a single structural skeleton:

Prior formulation	Recovered as	Reference
Schrödinger’s aperiodic crystal	R1 + R2 (extensible recognition)	(Schrödinger, 1944)
Szathmáry’s unlimited heredity	$C_S(L) = \Omega(L)$ (capacity theorem)	(Szathmáry, 2000)
Eigen’s error threshold	Dynamical bound on Q_L given R4, R5	(Eigen, 1971; Eigen et al., 1981)
Wagner’s neutral networks	Topological condition on R4 drift in Σ	(Wagner, 2005, 2011)
Pross’s dynamic kinetic stability	Thermodynamic-driving condition on R4	(Pross, 2011)
Hull’s replicator ontology	R3 (substrate reads its own product)	(Hull, 1980)
Maynard Smith’s units of selection	R3 + R4 jointly	(Maynard Smith, 2000)
Hofmeyr’s (F, A) -systems	R3 in categorical form	(Hofmeyr, 2017)
Vasas et al. GARD refutation	Capacity-R2 failure for composition substrates	(Vasas et al., 2010)
Walker–Davies top-down causation	R3 in causal-flow form	(Walker and Davies, 2013)

The substantive contribution beyond these prior formulations is twofold: (i) the separation of *capacity* (R1, R2, R3) from *generation* (R4), which makes precise why a perfect copier transmits but does not innovate, and (ii) the explicit scope-defining role of R5, which marks the boundary between intrinsic catalytic chemistry and externally optimized search procedures.

Within this synthesis: Schrödinger’s aperiodic-crystal intuition is precisely R1 (alphabet entropy) and R2 (extensible recognition) jointly; Szathmáry’s unlimited heredity is exactly the capacity-theorem conclusion $C_S(L) = \Omega(L)$; Eigen’s error threshold is a dynamical bound *on* Q_L that becomes well-defined only once R4 holds and (R5) places the kernel in the intrinsic-physicochemistry domain; Wagner’s neutral-network requirement is the topological structure of drift that R4 produces; Pross’s dynamic kinetic stability supplies the thermodynamic driving under which R4 is sustainable; Hull’s replicator ontology requires R3 (the substrate’s product must be itself a substrate); Hofmeyr’s (F, A) -systems give the categorical formulation of R3; Vasas et al.’s GARD refutation independently establishes capacity-theorem R2 failure for composition-vector substrates via a dynamical-fidelity route; Walker–Davies top-down causation supplies the causal-flow content of R3 in the information-substrate-to-phenotype direction.

The framework’s value lies in the unification: each prior formulation addresses a particular condition or pair of conditions; the capacity/generation split shows precisely how the conditions combine and which biological chemistries fail which.

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